

Real-time Market Data Forecasting (As a part of ACTS Management System)

Functional Requirement Specification Document

27-01-2025

Functional Requirements Specification, Real-time Market Data Forecasting

Functional Requirements Document

Name of Document	Functional Requirement Specification Real-time Market Data Forecasting
Author	Siripuram Suchandra, Agna Sebastian, Ajitesh Sharma, David D M, Himansh Gupta
Description of Content	Functional, Technical and Operational Requirements for the Real-time Market Data Forecasting System
Reference	Market Data Analysis and Forecasting Requirements Document
Approved by	Prasoon P
Date of Approval	27-01-2025

Functional Requirements Specification, Real-time Market Data Forecasting

Document Version Control

Version Number	Version Date	Prepared by	Reviewed by	Particulars/ Description for Change	Remarks
01	27-01-2025	Group 3	Prasoon P	First Release	

Functional Requirements Specification, Real-time Market Data Forecasting

Abbreviations

[illegible]

Functional Requirements Specification, Real-time Market Data Forecasting

CONTENTS

Table of Contents

Real-time Market Data Forecasting.....	1
Functional Requirement Specification Real-time Market Data Forecasting	2
Functional, Technical and Operational Requirements for the Real-time Market Data Forecasting System	2
1 INTRODUCTION	1
1.1 Purpose.....	1
This document outlines the architectural and design details of the Real-time Market Data Forecasting system. It provides a blueprint for developers to implement functionalities such as real-time stock dashboards, technical indicators, and 3-day forecasts using Prophet.....	1
1.2 Background	1
1.3 Scope.....	1
1.4 References	2
1.5 Document Overview	2
FUNCTIONAL REQUIREMENTS	3
2.1 Data Streaming.....	3
2.1.1 Description	3
2.1.2 Functional Requirements Identified.....	3
2.1.3 Fields Validations	4
3.2.1.4 Pre-requisites, Assumptions and Dependencies	4
2 External Interfaces.....	4

1 INTRODUCTION

1.1 Purpose

This document outlines the architectural and design details of the Real-time Market Data Forecasting system. It provides a blueprint for developers to implement functionalities such as real-time stock dashboards, technical indicators, and 3-day forecasts using Prophet.

1.2 Background

The Real-Time Market Data Forecasting project aims to predict stock price movements of Nifty 50 companies. It leverages real-time data streamed through the yfinance API and hosted on AWS. This system is designed to assist users in making informed financial decisions by providing accurate predictions displayed on a website chart.

1.3 Scope

This project includes the following functionalities:

- Streaming real-time market data.
- Predicting stock prices using AI models.
- Visualizing actual vs. predicted prices on an interactive web platform.
- Automatically halting API calls after trading hours to optimize performance and cost.

The system targets traders, investors, and financial analysts seeking insights into stock price trends. It will be accessible via a web interface, ensuring secure and user-friendly interaction

Figure 1: System Architecture

1.4 References

- a) Market Data Analysis and Forecasting Requirements Document
- b) API Documentation for yfinance
- c) Kafka Integration Guidelines

1.5 Document Overview

This document gives an overview of the Real-Time Market Data Forecasting its interactions and functionality. It also describes the type of stakeholders that use the system and what functionalities are available for each type. The following FRS contains the details of <what the project is doing>. It provides the detailed functions of the model with permitted Constraints, Assumptions, Dependencies and Requirements subsets.

FUNCTIONAL REQUIREMENTS

List the functionalities of the system.

2.1 Data Streaming

2.1.1 Description

The system streams real-time stock market data from the **yfinance API** and integrates it into the forecasting model through Kafka.

2.1.2 Functional Requirements Identified

Functionality 1			
SN.	Functionality	Process	Remarks/ Additional Info
FR 1.1	Train predictive models	Use historical data to train AI models for accuracy.	Models should be retrained periodically.
FR1.2	Forecast stock prices	Predict future prices and update outputs in real-time.	Support multi-threaded processing.

2.1.3 Fields Validations

This table lists the various fields that will be there in the form for the requirement

S N	Field Name	Field Description	Validations	Remarks
1	Name of field	Field description	● Validations	Mandatory/non mandatory

3.2.1.4 Pre-requisites, Assumptions and Dependencies

Any prerequisite is there or any dependency. (Example user data should be uploaded or PRN should be generated)

2 External Interfaces

Explain if any external interface is to be connected like redirection to some other portal or information fetch from some other portal.